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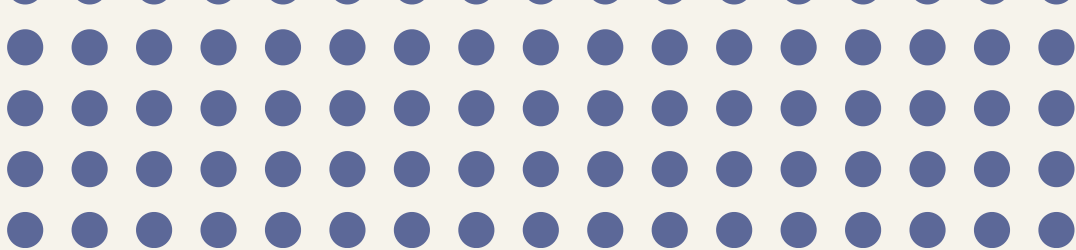
University of
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FINAL YEAR PROJECT

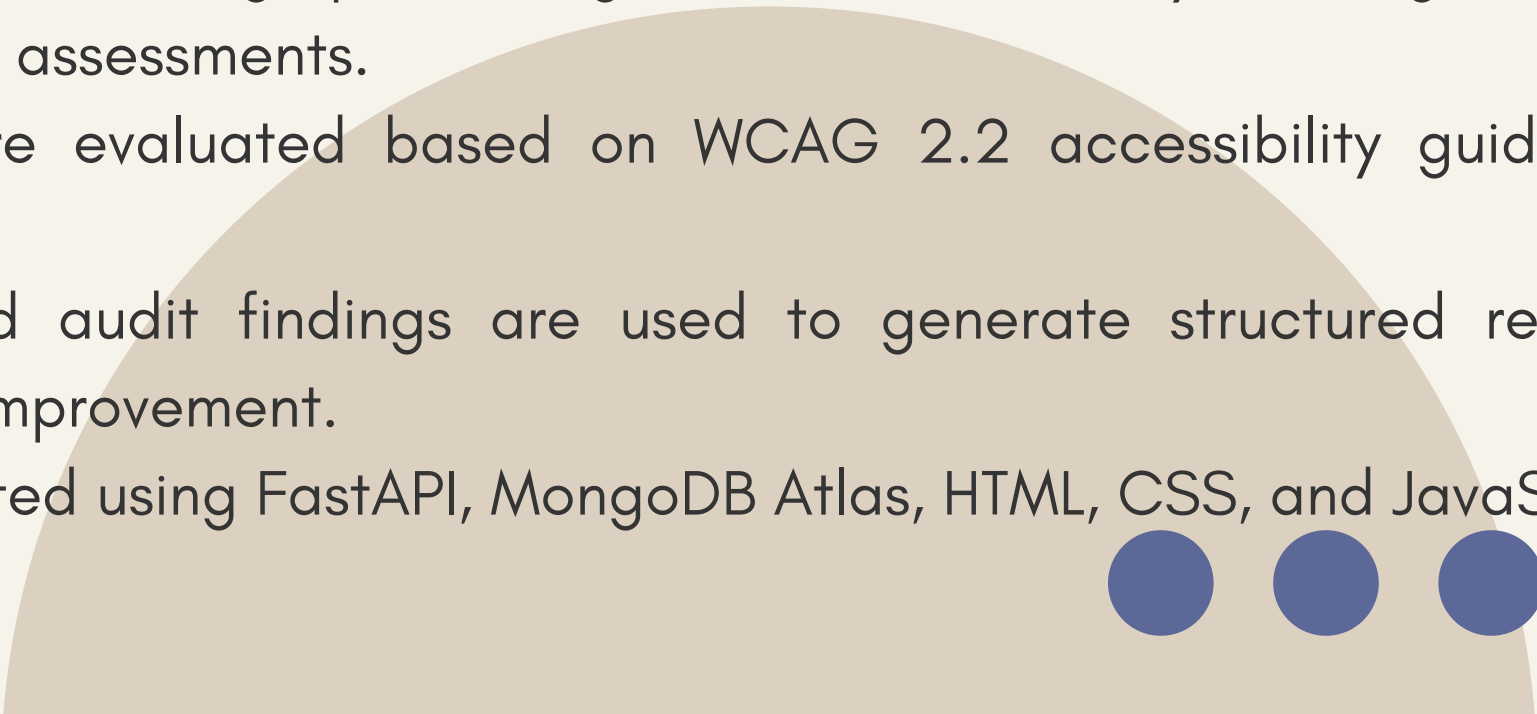
AI CHATBOT FOR UI IMPROVMENT

Kenneth Chong Vui Kin – SCSJ2100435

Supervisor– Mr Abdul Arriff Abdul Jalal



Abstract

- UI/UX quality impacts user satisfaction and engagement.
 - Manual UI evaluations are time-consuming and subjective.
 - Early usability issue detection is important but challenging.
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- Developed an AI Chatbot for UI Improvement that automates UI/UX evaluation through screenshot analysis and conversational feedback
 - Traditional usability evaluation methods are often time-consuming, subjective, and dependent on expert expertise.
 - The system combines OpenCV image processing, rule-based usability auditing, and Google Gemini 2.5 Flash to perform comprehensive UI assessments.
 - Uploaded screenshots are evaluated based on WCAG 2.2 accessibility guidelines and Nielsen's usability heuristics.
 - Extracted UI metrics and audit findings are used to generate structured recommendations for usability, accessibility, and design improvement.
 - The system was implemented using FastAPI, MongoDB Atlas, HTML, CSS, and JavaScript.
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Problem Statement

Time-Consuming and Inconsistent Evaluations

- Manual heuristic evaluations require significant time and UI/UX expertise
- Findings vary between evaluators due to subjective judgment and experience

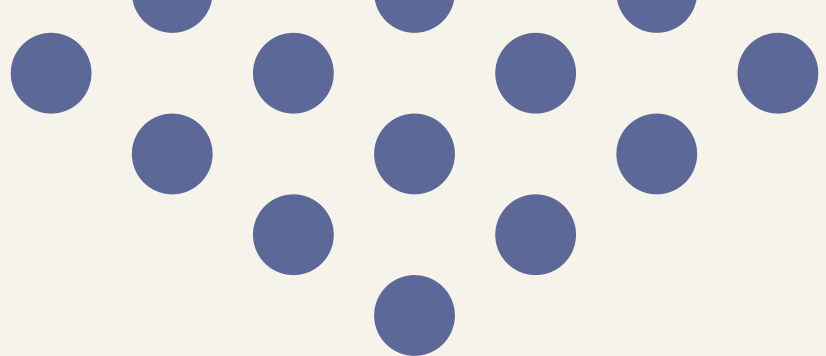
Complexity of Modern Applications

- Modern applications contain multiple features, screens, and user interactions
- Manually reviewing every screen and workflow is impractical

Late Detection of Usability Issues

- Lack of immediate feedback causes issues to be discovered late in development
- Fixing problems in later stages requires more time and cost





Objectives

1

Develop a tool that analyzes UI screenshots/descriptions and provides usability improvement suggestions

2

Implement a rule-based engine to detect common UI/UX issues consistently

3

Integrate an AI model to deliver enhanced usability feedback during the design phase

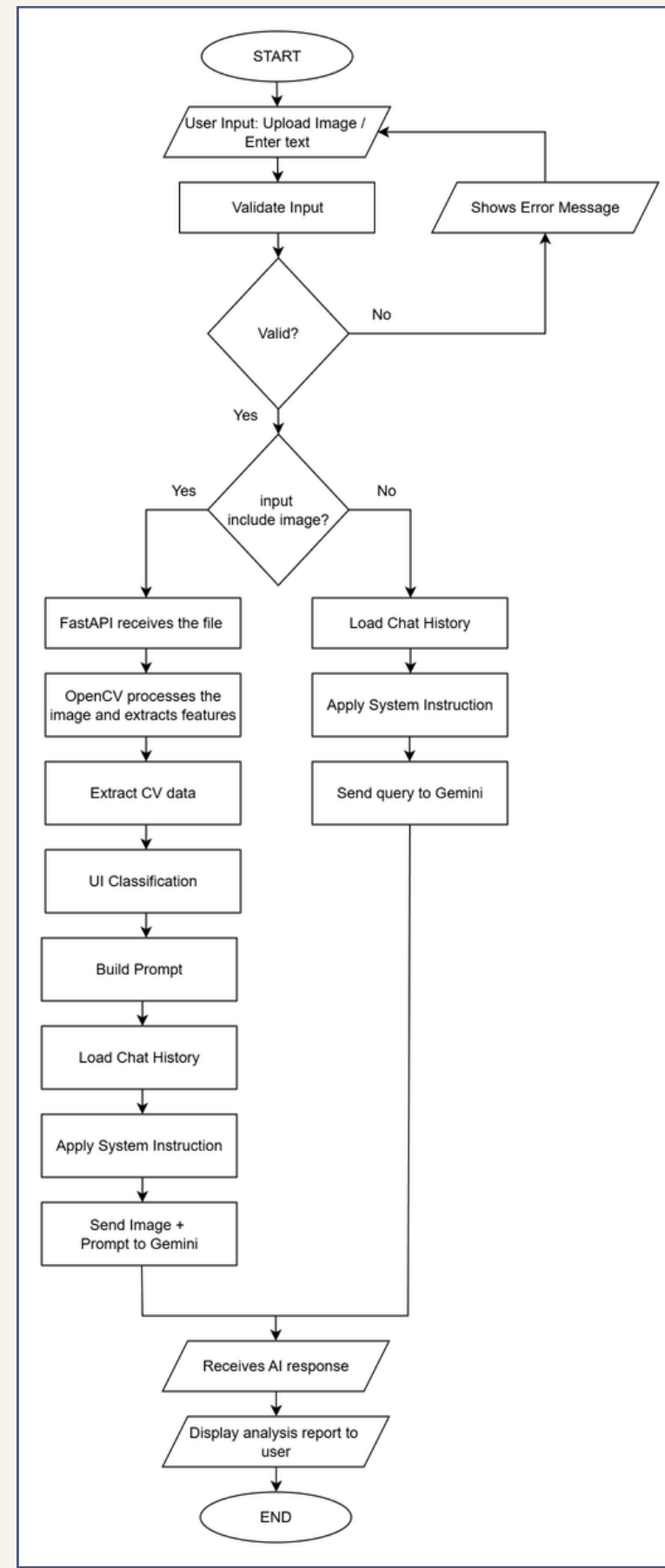


Methodology



System Design

System Flow Chart



Project Outcome

UI Audit AI



Technical Metrics

- **Whitespace:** 49.62%
- **Small Elements:** 26

Point 1: Insufficient Color Contrast

Finding: Several text elements and interactive components across the page demonstrate insufficient color contrast, failing to meet WCAG AA guidelines. Noteworthy examples include the placeholder text in the search bar, the phone number and related account links in the header, and the white text on the main call-to-action buttons ("Shop Light Ups" and "Custom Blinkiness"). This issue significantly impacts readability for all users, particularly those with visual impairments, and detracts from the overall (Aesthetic) quality.

Severity: Major

Practical Fix: Adjust the color palette to ensure all text, placeholder text, and essential UI components meet a minimum contrast ratio of 4.5:1 (or 3.0:1 for large text and non-text contrast for UI components) against their

Ask about your UI design...



AI can make mistakes. Check important info.



Limitations

AI Response Latency and Inconsistency

- Gemini 2.5 Flash processing introduces noticeable response delays due to AI reasoning and recommendation generation
- The AI model cannot answer the question when it is experiencing high demand usage

Lack of UI Screenshot Validation

- The system does not verify whether uploaded images are UI screenshots, allowing non-UI images (e.g., selfies or photos)

Manual Verification Requirement

- Human verification is still needed to confirm the accuracy of certain AI-generated findings and severity classifications
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Future Enhancement



Advanced UI Component Recognition

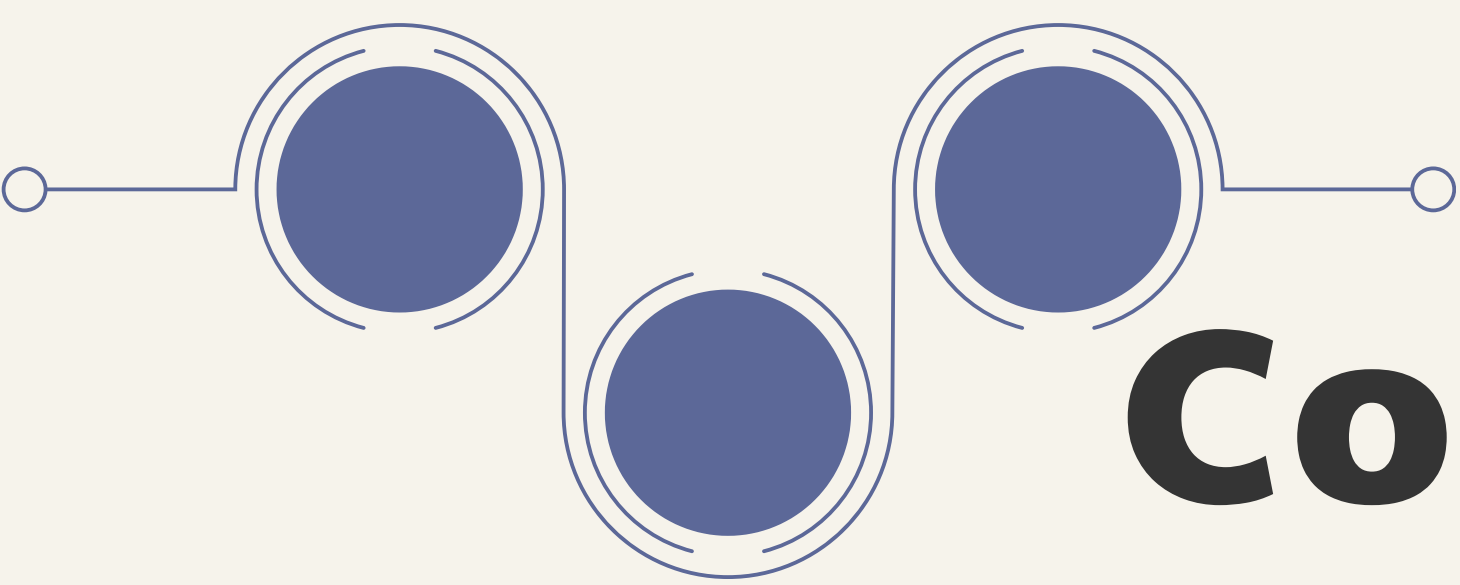
- Future improvements can use deep learning-based object detection to enhance accuracy in identifying UI elements

UI Screenshot Validation Module

- A preprocessing step can be added to detect valid UI screenshots and filter out non-UI images before analysis

Performance Optimization

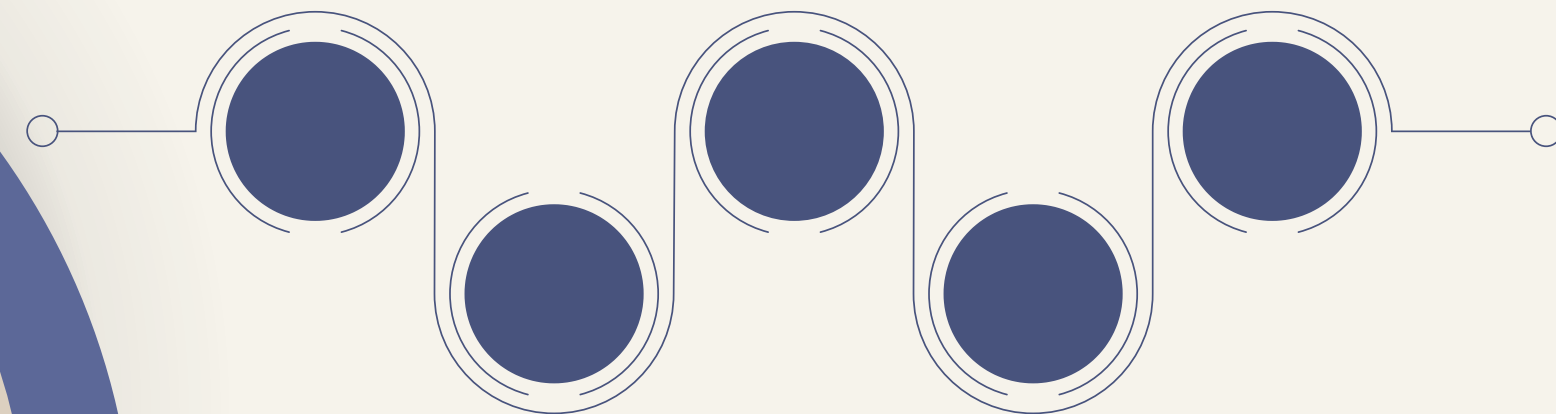
- System response time can be improved through caching mechanisms and optimized AI request processing pipelines



Conclusion

This project successfully designed, developed and evaluated an “AI Chatbot for UI Improvement” that combines rule-based usability auditing, computer vision analysis and generative AI reasoning.

The system integrates OpenCV-based image analysis, WCAG 2.2 accessibility standards, Nielsen’s heuristics and the Google Gemini 2.5 Flash model to provide automated UI/UX evaluation through a chatbot interface. The proposed hybrid architecture successfully detected usability issues and generated findings and recommendations.



Poster

AI CHATBOT FOR UI IMPROVEMENT

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ABSTRACT

This project presents an AI Chatbot for UI Improvement that analyzes UI screenshots using OpenCV, rule-based usability auditing, and Google Gemini 2.5 Flash. Based on WCAG 2.2 and Nielsen's heuristics, the system identifies usability issues and generates improvement recommendations. Results demonstrate fast, consistent, and actionable feedback for UI/UX evaluation.

INTRODUCTION

UI and UX quality greatly influence user satisfaction and the success of digital products. However, traditional usability evaluations are often time-consuming, subjective, and dependent on expert reviewers. These limitations can delay the detection of usability issues and increase development costs. Therefore, an automated and accessible approach is needed to support efficient UI/UX evaluation.

OBJECTIVES

- To Develop a tool that analyzes UI screenshots and provides usability improvement suggestions
- Implement rule-based auditing to detect common UI/UX issues consistently
- To integrate Gemini 2.5 Flash to generate enhanced usability feedback and recommendations

USE CASE DIAGRAM



FRAMEWORK



METHODOLOGY



FUTURE ENHANCEMENT

- Improve AI accuracy through advanced UI recognition and multimodal analysis
- Enhance the system with user accounts, analytics dashboards, and faster processing
- Add multi-language support, visual redesign suggestions, and accessibility compliance reports

CONCLUSION

This project developed an AI Chatbot for UI Improvement that combines rule-based auditing, OpenCV, and Google Gemini 2.5 Flash for automated UI/UX evaluation. The system successfully identified usability issues and generated actionable recommendations, demonstrating the effectiveness of a hybrid AI approach for design evaluation.

PROJECT OUTCOME

